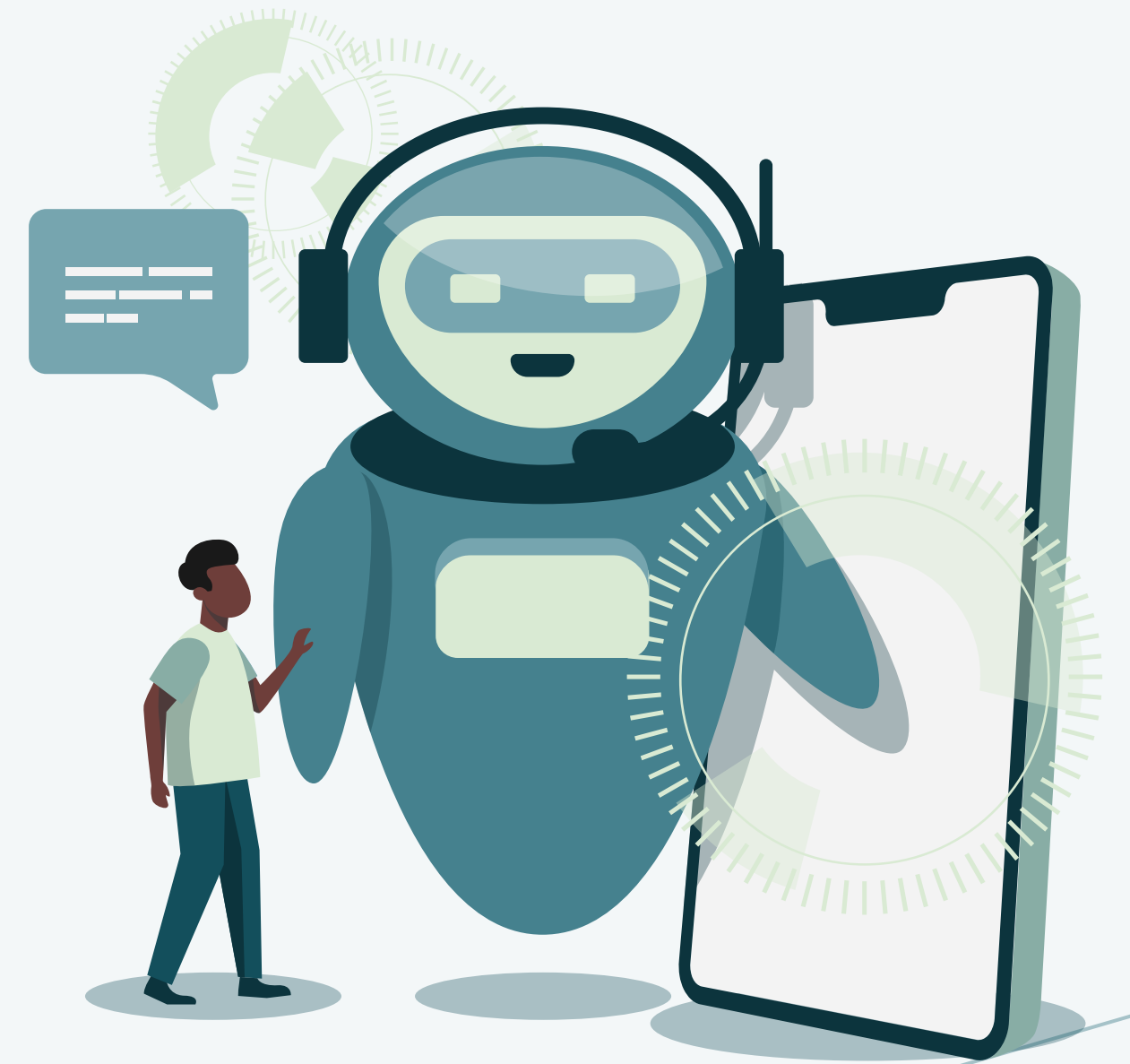


“What if every company had a digital HR team — that never slept, never got biased, and never missed the best candidate?”



HireAI - A Multi-Agent Hiring System

Revolutionizing Technical
Recruitment with a Multi-Agent
Workforce

Team: Rayyan, Deepesh, Jewel, Vaibhav
Mentor: Dr. Dheeraj
Module: Agentic AI | 2025



We didn't build a chatbot — we built an AI Hiring Tool

From Idea → Intelligence (Our Journey)

Week 1	Researched HR inefficiencies, validated market gap
Week 2	Developed prototype orchestration using CrewAI
Week 3 - 4	Integrated Whisper voice interviews + FAISS embedding engine
Week 5	Functional 7-agent HR automation pipeline with dashboard

From concept to a working multi-agent Hiring system in 5 weeks.





Agenda

- Problem & Vision
- Research & Market Opportunity
- HireAI Solution Overview
 - System Architecture & Agents
 - Key Features & Results
 - Roadmap & Future Scope
 - Team & Conclusion

The Broken Reality of Hiring

**75% of resumes are never
seen by humans**

(Forbes, 2024)

**52% of recruiters say
screening is the most time-
consuming step**

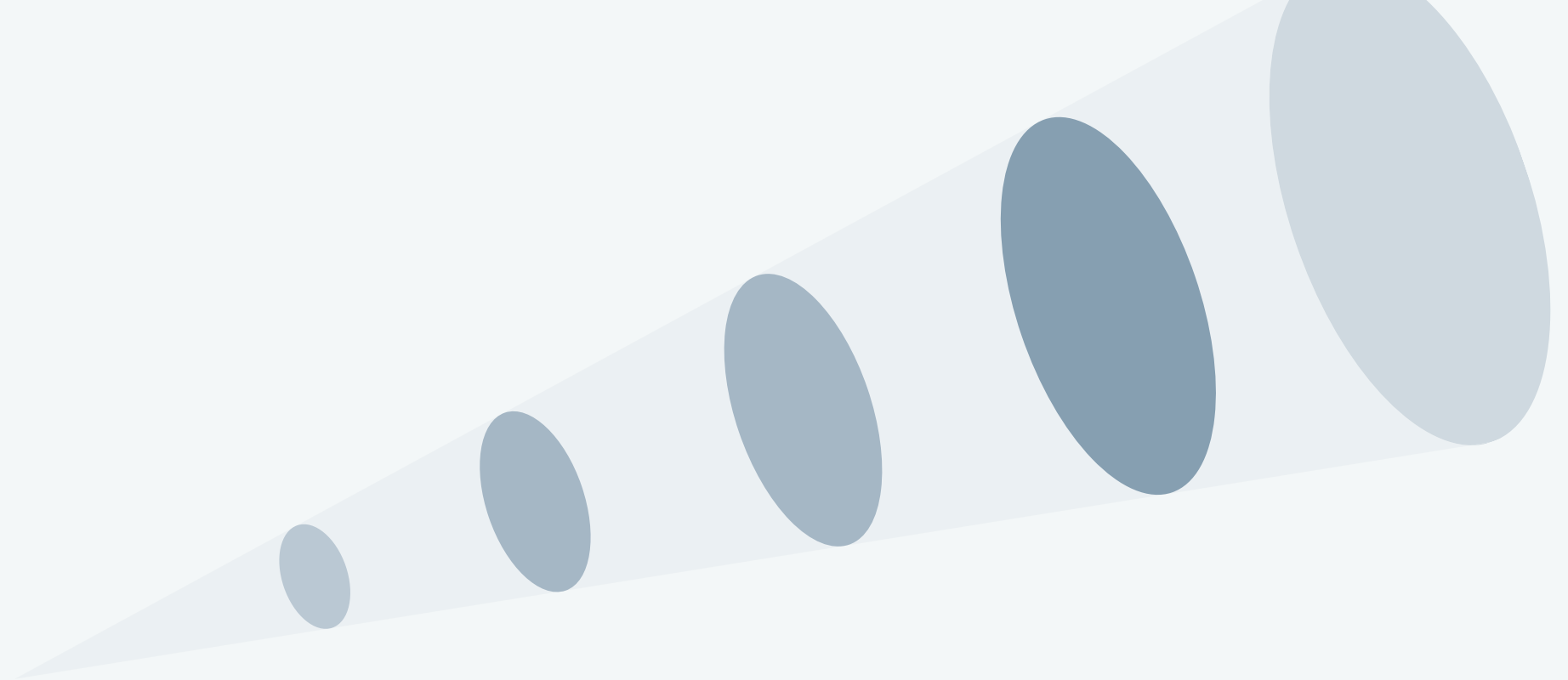
(LinkedIn Talent Trends)

**Bias and inconsistency cost
companies \$3.1B annually**

(SHRM Report, 2025)



**Hiring is inefficient, expensive, and
biased — HireAI fixes all three.**



Market & Opportunity - A Multi-Billion Dollar Problem

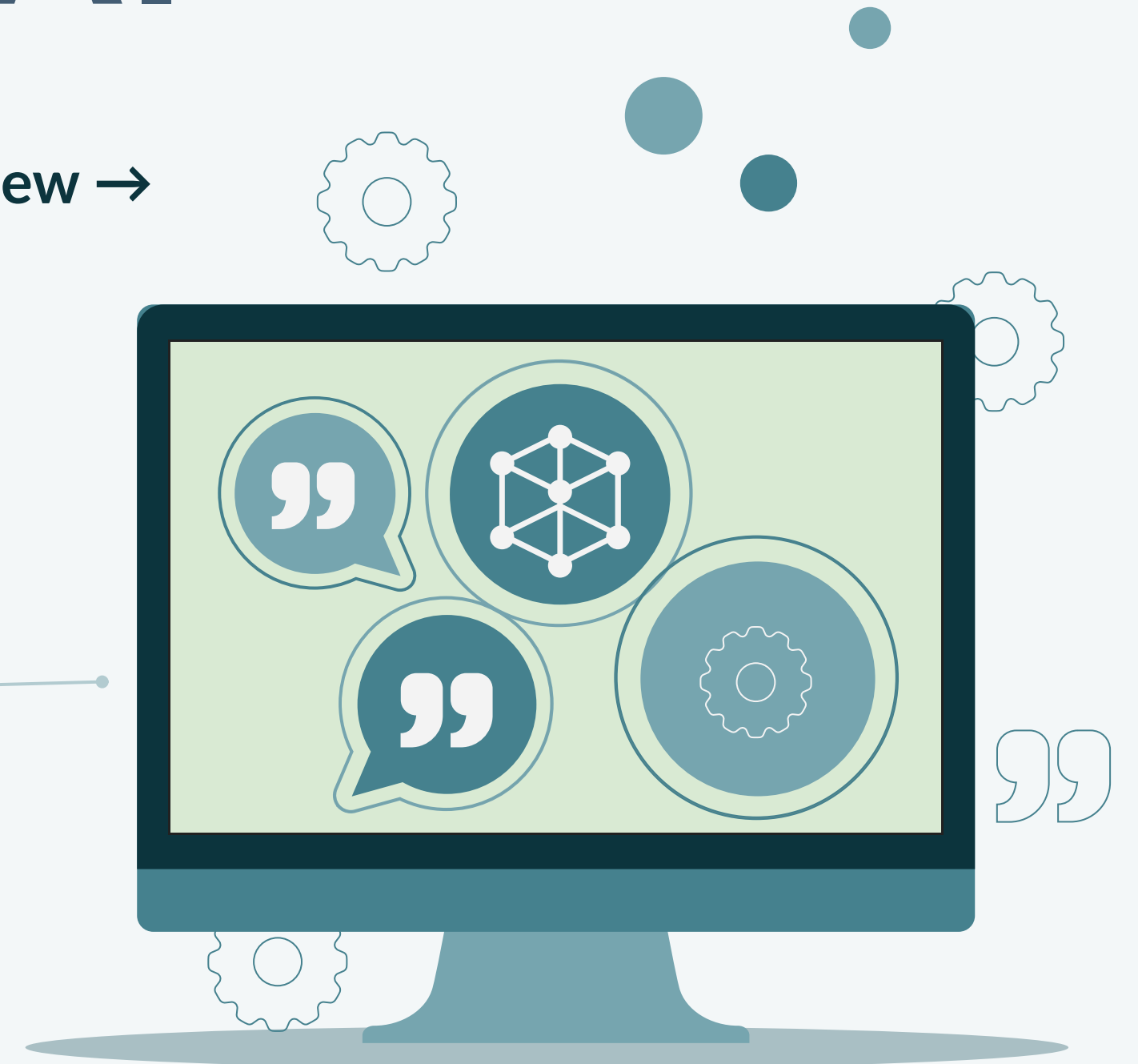
- Global HR Tech Market: USD 36.6B by 2030 (Grand View Research, 2025)
- High Demand: Companies are desperate for tools that reduce hiring costs and improve the quality of hires.
- The Compliance Imperative: With increasing regulation around fair hiring (DEI, EEOC), automated compliance is no longer optional.
- At 25% CAGR, HR + AI is one of the top-3 global growth verticals — yet 60% of SMEs still rely on manual screening (NASSCOM-BCG)

Our Solution: HireAI

HireAI is An Autonomous Hiring Platform that automates the full recruitment lifecycle — from Resume Parsing → Matching → Interview → Evaluation → Offer.

Powered by: CrewAI, LangChain, Groq LLM, Whisper, FAISS, MongoDB

“We replaced fragmented HR tasks with a synchronized AI workforce.”



What Our System Can Do

Autonomous AI Interviews:
Conducts real-time,
proctored conversational
interviews.

Intelligent Matching: Scores
candidates with high
accuracy using a two-stage
(FAISS + LLM) process.

Proactive Sourcing:
Automatically re-engages
past candidates for new
roles.

Parallel Vetting: Enriches
profiles by analyzing GitHub
and public web presence
simultaneously.

End-to-End Automation:
Manages everything from
application to final email.



From Raw Data to Actionable Insights

The system doesn't just process data; it produces clear, actionable outputs for recruiters.

**Ranked Candidate
Lists**

**Detailed Interview
Reports**

Compliance Logs

Enriched Profiles



Meet the Crew: The Agent - 1

The Orchestrator Agent: The Project Manager

- **Role:** Recruitment Process Orchestrator
- **Function:** This agent is the manager and central brain of the entire system. It does not perform tasks itself; it directs the workflow.

Initiates Workflow

Delegates Tasks

Makes Key Decisions

Manages the Process





Meet the Crew: The Agent - 2

The Resume Parsing Agent: The Document Analyst

- **Role: Document Processing Specialist**
- **Function: This agent is an expert at reading and understanding unstructured documents.**

Extracts Data

Generates Embeddings

Stores Information



Meet the Crew: The Agent - 3

The Compliance Agent: The Ethics Officer

- **Role: AI Ethics & Compliance Monitor**
- **Function: This agent acts as an automated legal and ethical advisor, running as a mandatory step in the screening process.**

Scans for Bias

Assesses Risk

Provides Recommendations

Creates an Audit Trail



Meet the Crew: The Agent - 4

The GitHub Agent: The Code Reviewer

- **Role: GitHub Profile Analyst**
- **Function: This parallel agent acts like a senior engineering manager, providing an evidence-based assessment of a candidate's technical skills.**

Analyzes Repositories

Evaluates Project Quality

Measures Activity



Meet the Crew: The Agent - 5

The Public Search Agent: The Research Analyst

- **Role: Public Presence Analyst**
- **Function: This parallel agent scours the web to discover a candidate's footprint in the wider technical community.**

Finds Technical Content

Delegates Tasks

Makes Key Decisions



Meet the Crew: The Agent - 6

The Synthesizer Agent: The Profile Analyst

- **Role: Senior Recruiting Analyst**
- **Function: This agent is the manager and central brain of the entire system. It does not perform tasks itself; it directs the workflow.**

Reads All Inputs

Creates a Narrative

Delivers a Coherent
Summary



Meet the Crew: The Agent - 7

The Matching Agent: The Expert Recruiter

- **Role:** Intelligent Matching Specialist
- **Function:** This agent is the core of the screening process, determining a candidate's suitability for open roles.

Performs Vector Search

Conducts LLM Evaluation

Generates a Score



Meet the Crew: The Agent - 8

The Sourcing Agent: The Talent Scout

- **Role: Proactive Talent Sourcer**
- **Function: This agent turns your database of past applicants into a valuable, living talent pool.**

Monitors New Jobs

Searches the Past

Filters and Qualifies

Re-engages Talent



Meet the Crew: The Agent - 9

The Interview Agent: The Autonomous Interviewer

- **Role: Real-Time Conversational Interviewer**
- **Function: This agent acts as a fully autonomous interviewer, capable of conducting intelligent, dynamic, and bias-free conversations with candidates.**

Conducts Live Interviews

Dynamic Questioning

Speech Intelligence

Data Integration



Meet the Crew: The Agent - 10

The Communication Agent: The Candidate Liaison

- **Role: Candidate Experience & Notification Specialist**
- **Function: This agent manages all communications between the hiring system and candidates, maintaining a consistent and professional experience.**

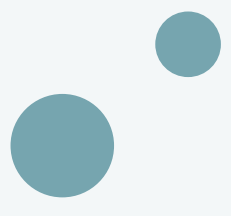
Automated Correspondence

Personalization Engine

Multi-Channel Integration

Status Tracking

Human-in-Loop Option



Innovation Highlight 1 — The Compliance Agent



1) Core Idea:
Ethics built into automation.

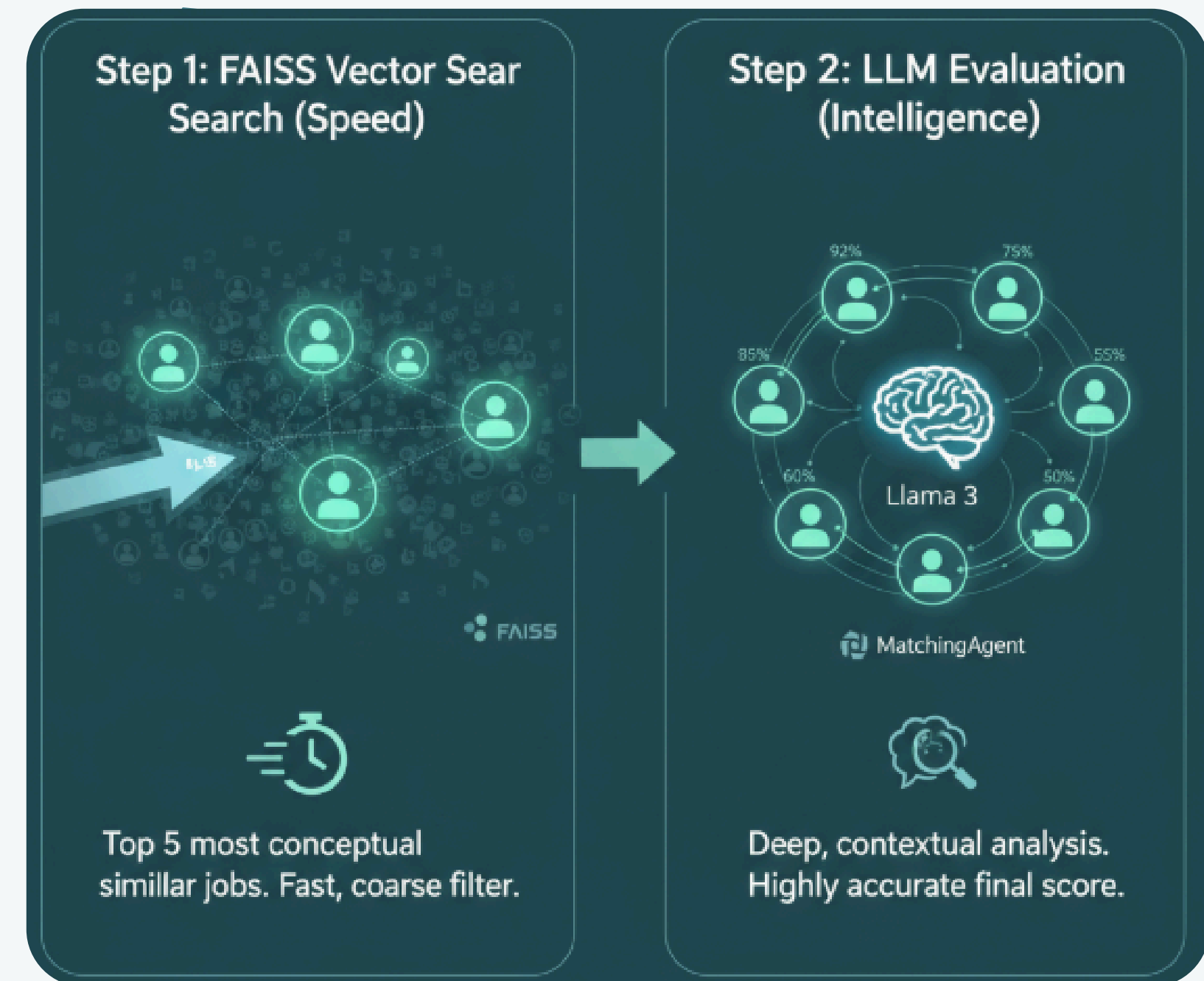


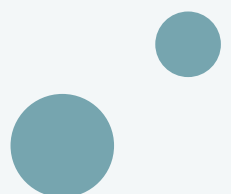
2) What It Does:
Detects and masks bias-triggering data.
Ensures blind, fair candidate evaluation.
Logs every step for transparency.

3) Impact:
Turns AI hiring from efficient → ethical.

Innovation Highlight 2 - The Two-Stage Match - Combining Speed and Intelligence

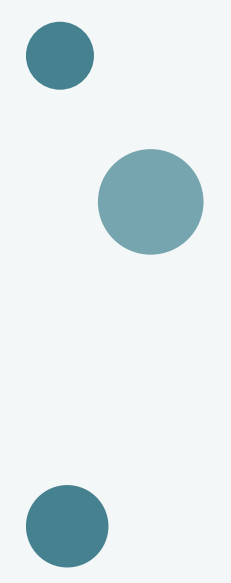
- **Step 1: FAISS Vector Search (Speed):**
The system instantly finds the Top 5 most conceptually similar jobs from a database of thousands. This is the fast, coarse filter.
- **Step 2: LLM Evaluation (Intelligence):**
The MatchingAgent then asks Llama 3 to perform a deep, contextual analysis on only these top 5 jobs, resulting in a highly accurate final score.

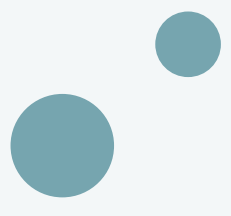




Innovation Highlight 3 - Going Beyond the Resume

The Orchestrator delegates to the VettingCrew, which works in parallel to save time.

- Public Search Agent: Finds blogs, articles, and talks.
 - GitHub Agent: Scans code for languages, quality, and activity.
 - Synthesizer Agent: Merges the findings into one "Enriched Profile."
 - Benefit: Provides a deep, 360-degree view of a candidate's skills and passion, not just what's on their resume.
- 



Technical Architecture

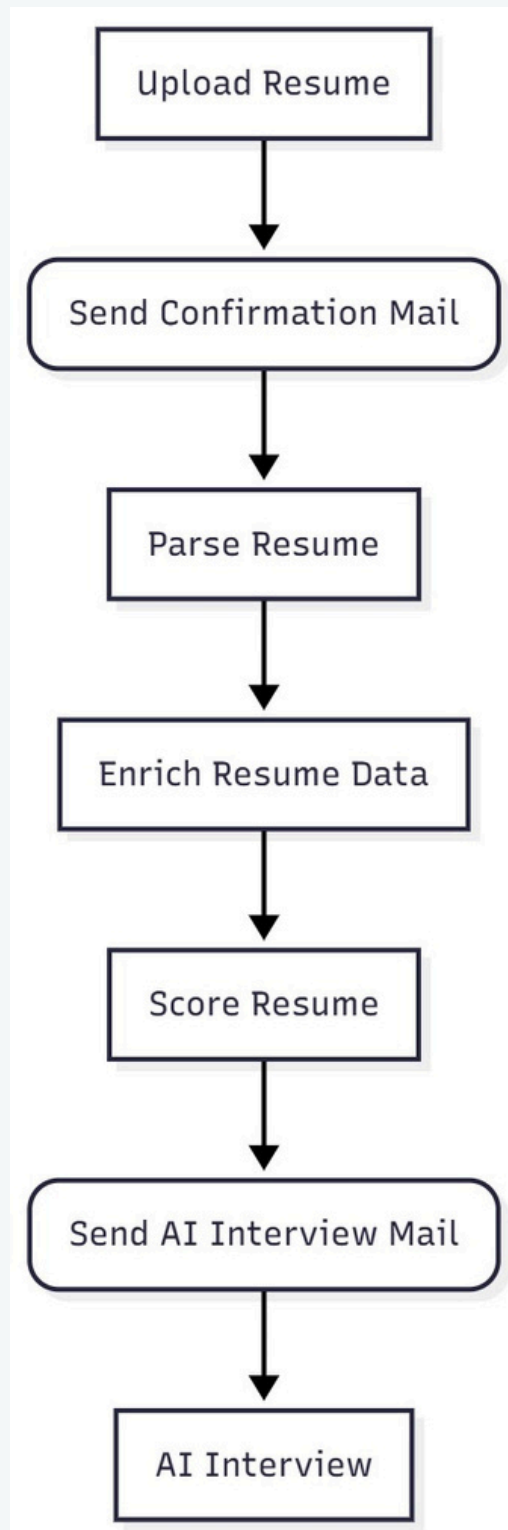
Core Stack:

- **Frontend:** React, Google Forms, WebSocket
 - **Backend:** FastAPI, CrewAI Orchestrator, LangChain
 - **AI Layer:** Groq LLM, OpenAI LLM, Whisper, TTS API
 - **Storage:** MongoDB + FAISS Vector DB
 - **Integrations:** MCP Protocol, EmailTool
- 

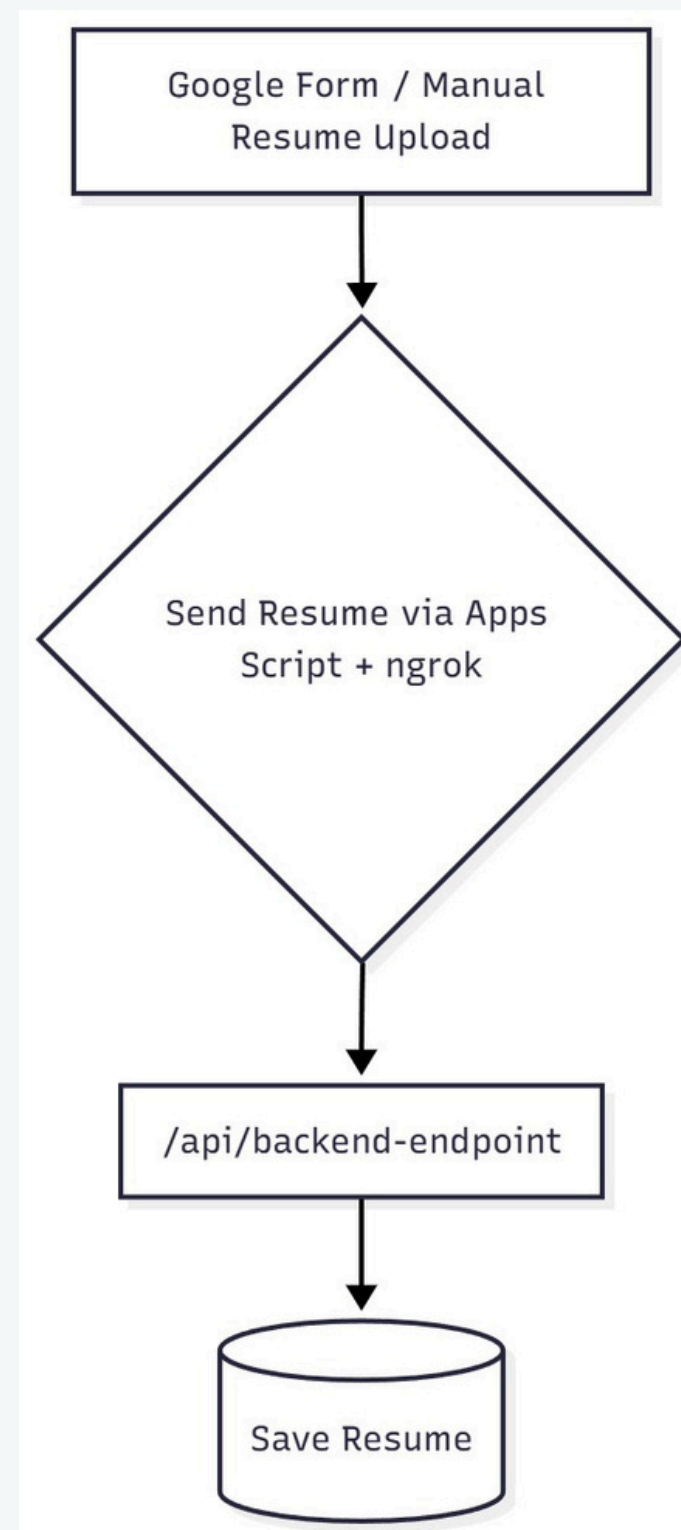
CrewAI enables concurrent multi-agent workflows with shared memory and error recovery.

System Architecture Overview

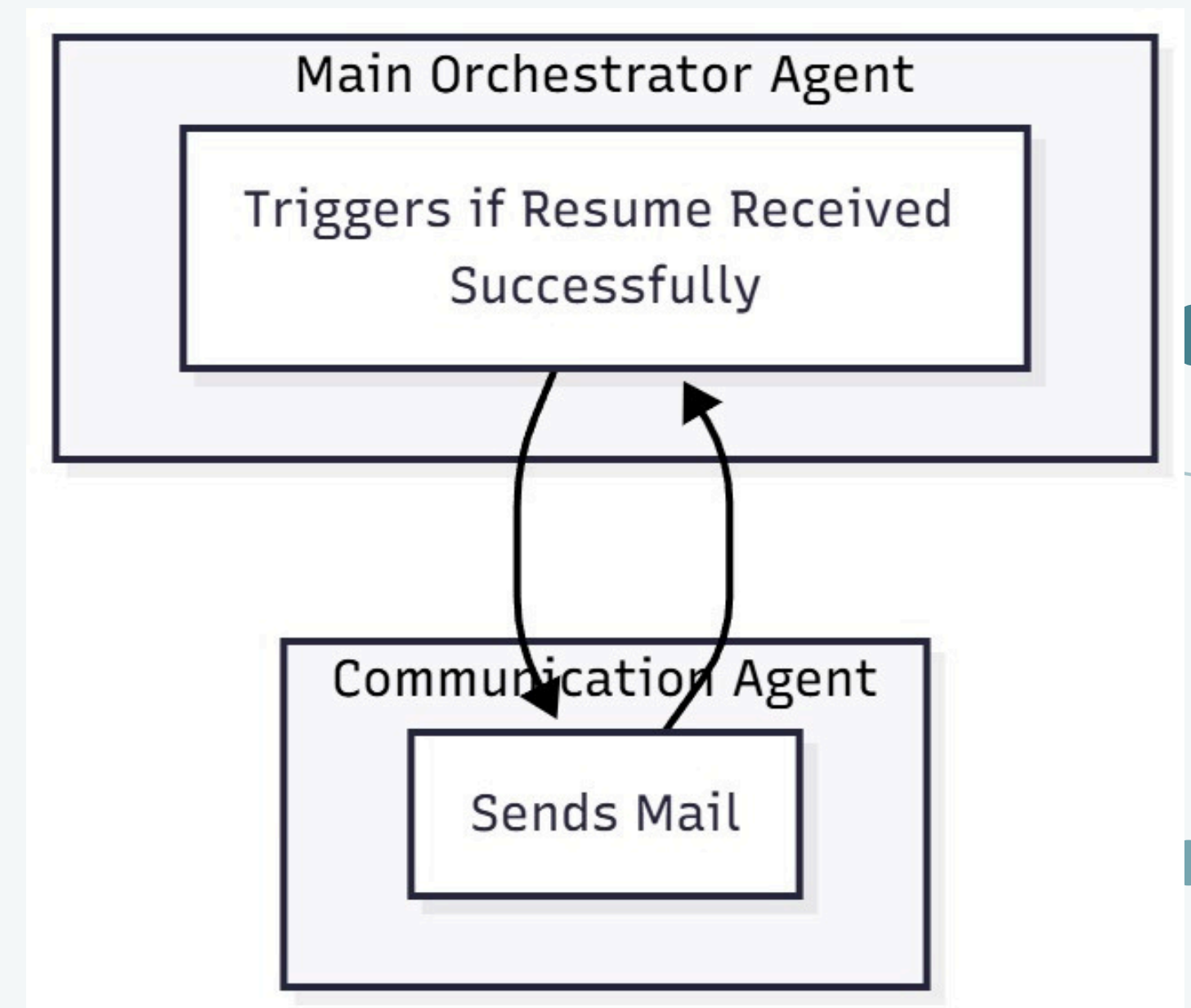
Micro Architecture Flow - 1



Overview tech flow chart



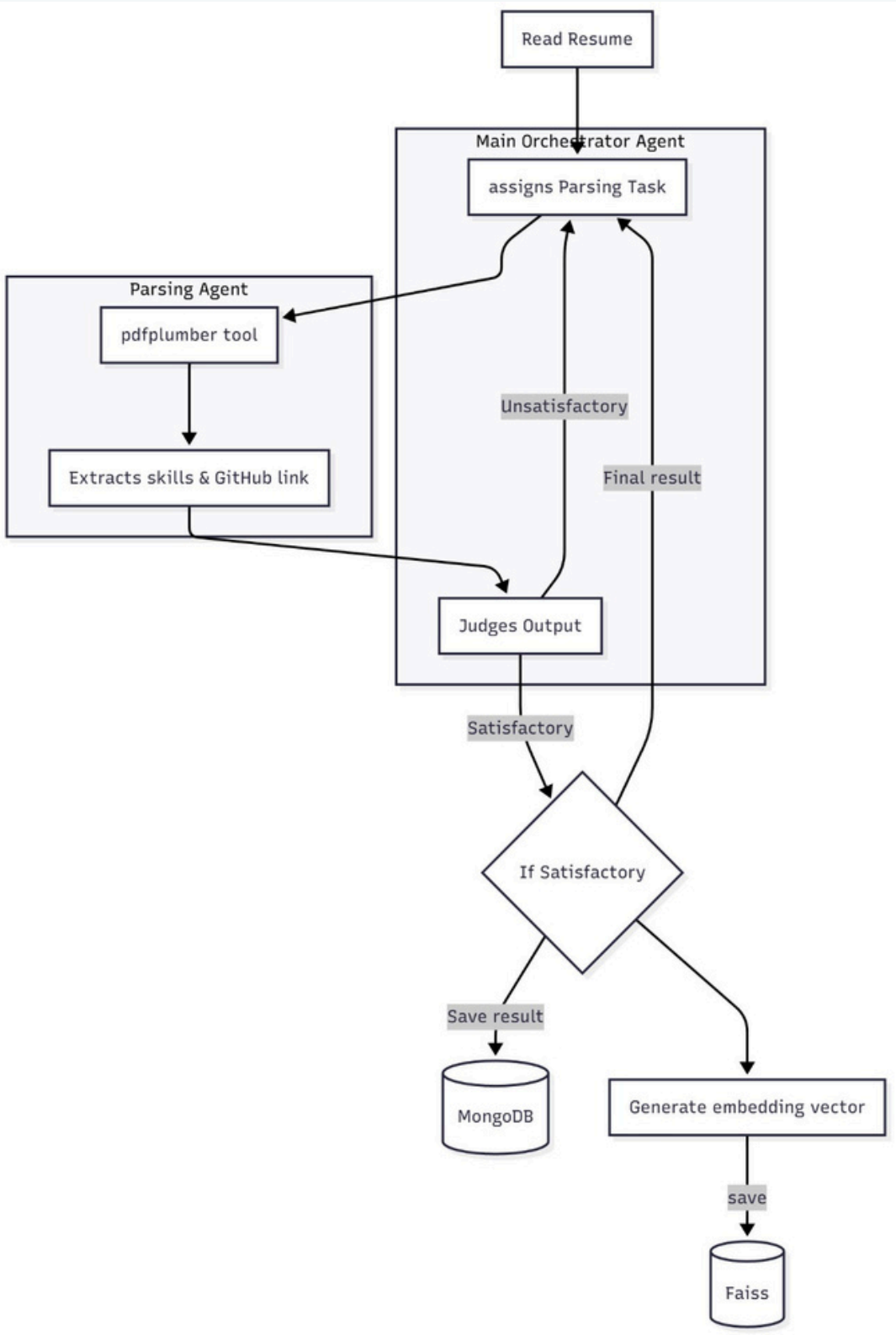
Upload resume flow chart



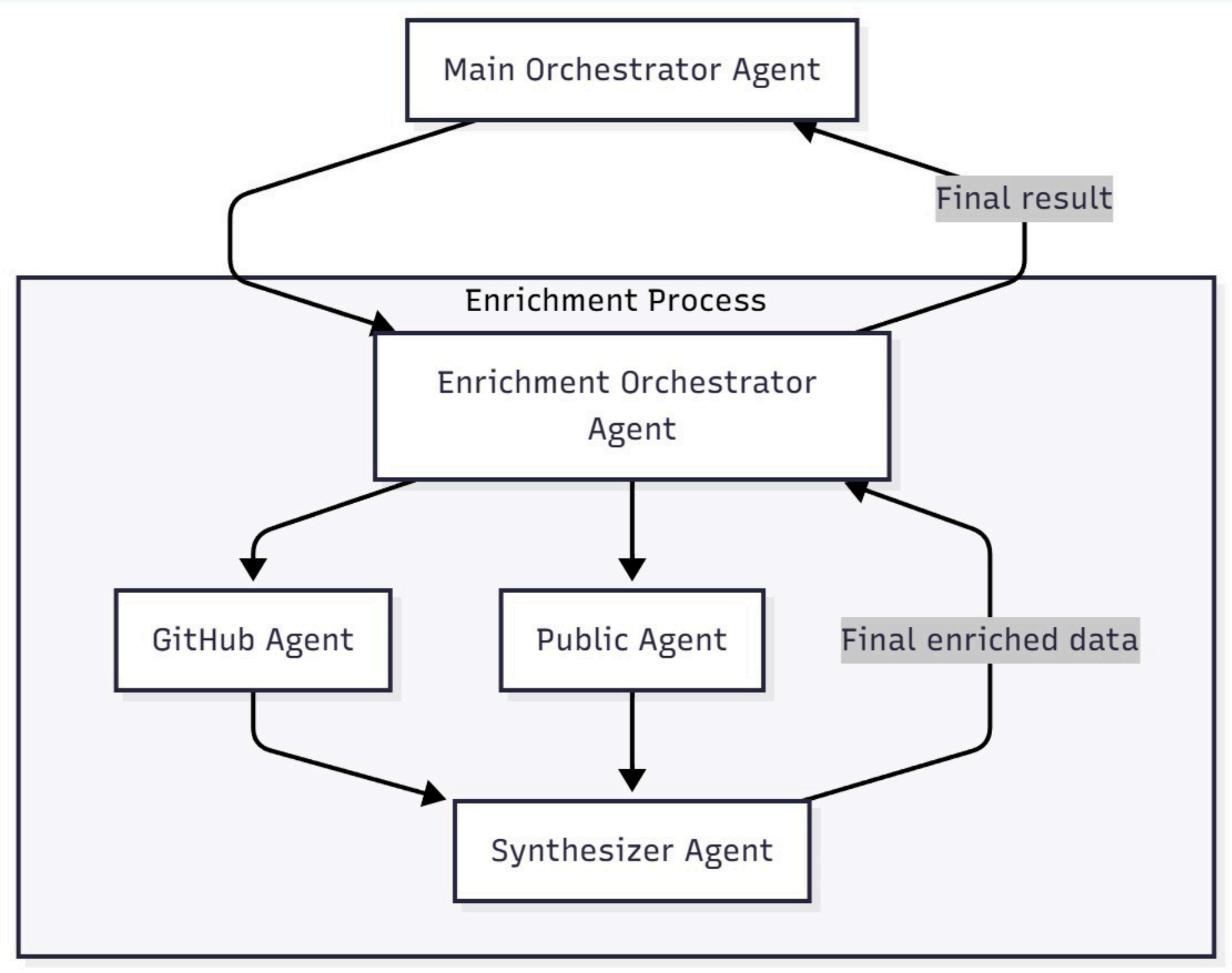
Send received mail flowchart

System Architecture Overview

Micro Architecture Flow - 2



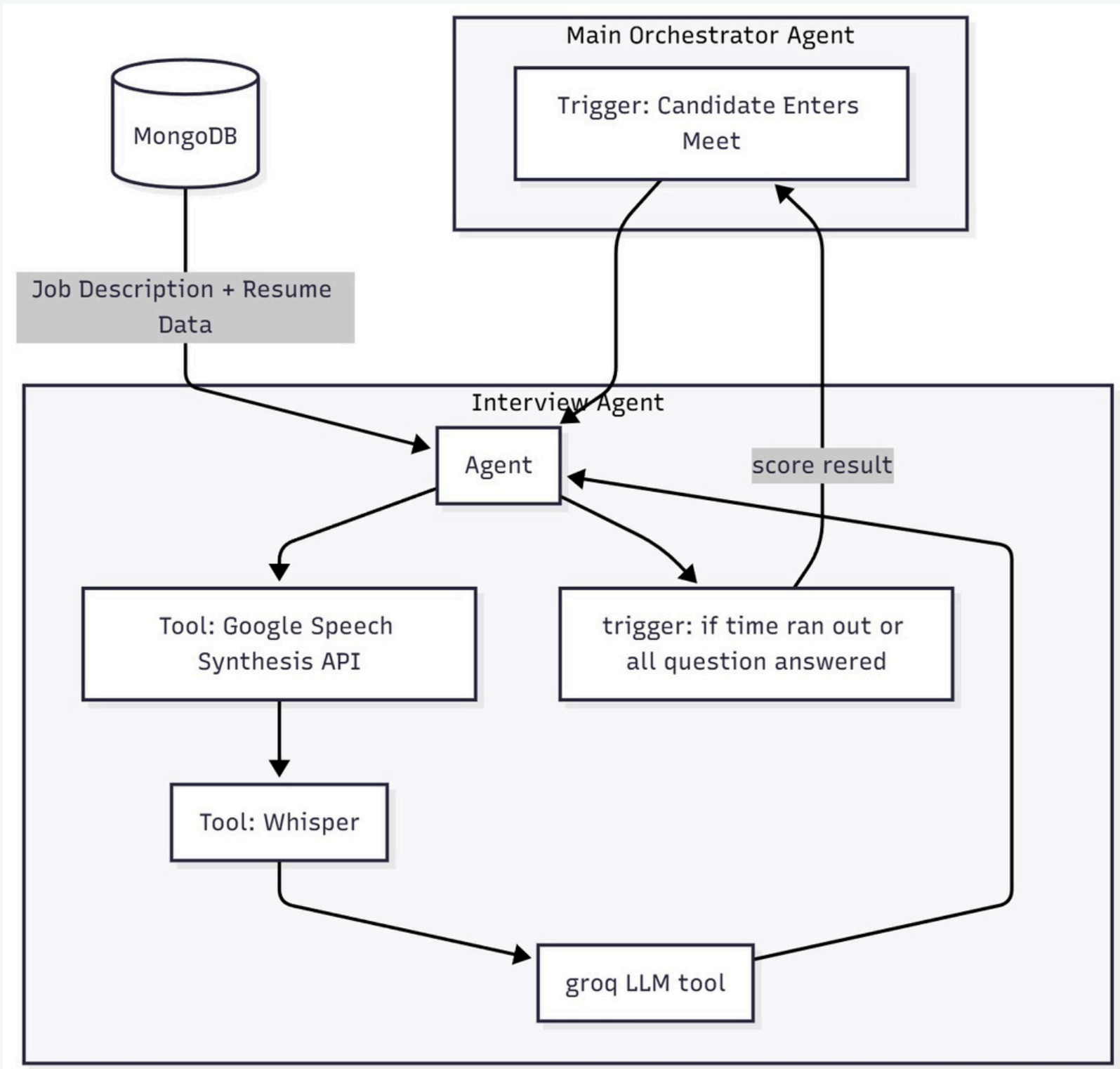
Parse resume flow chart



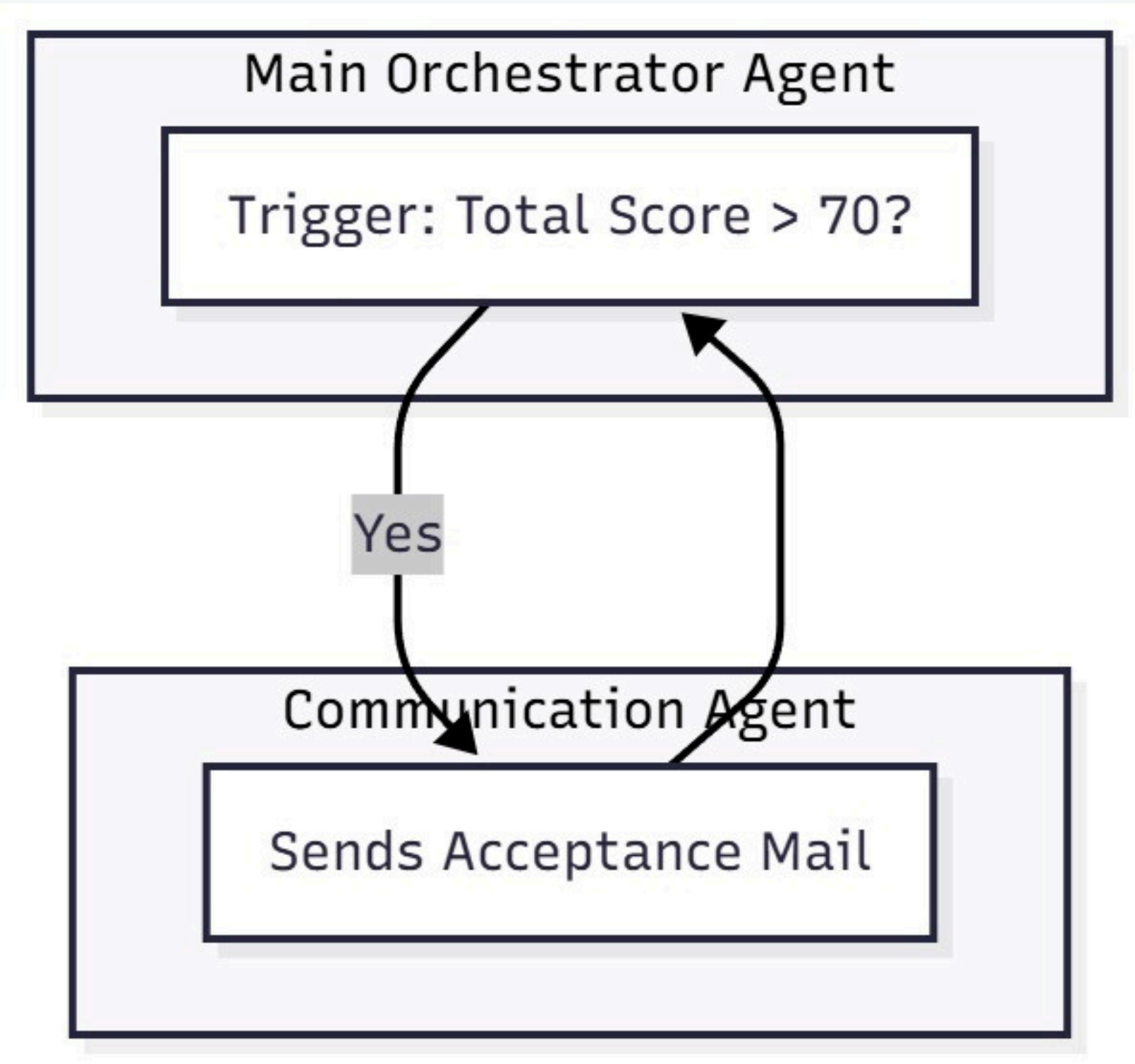
Enrich resume flow chart

System Architecture Overview

Micro Architecture Flow - 4

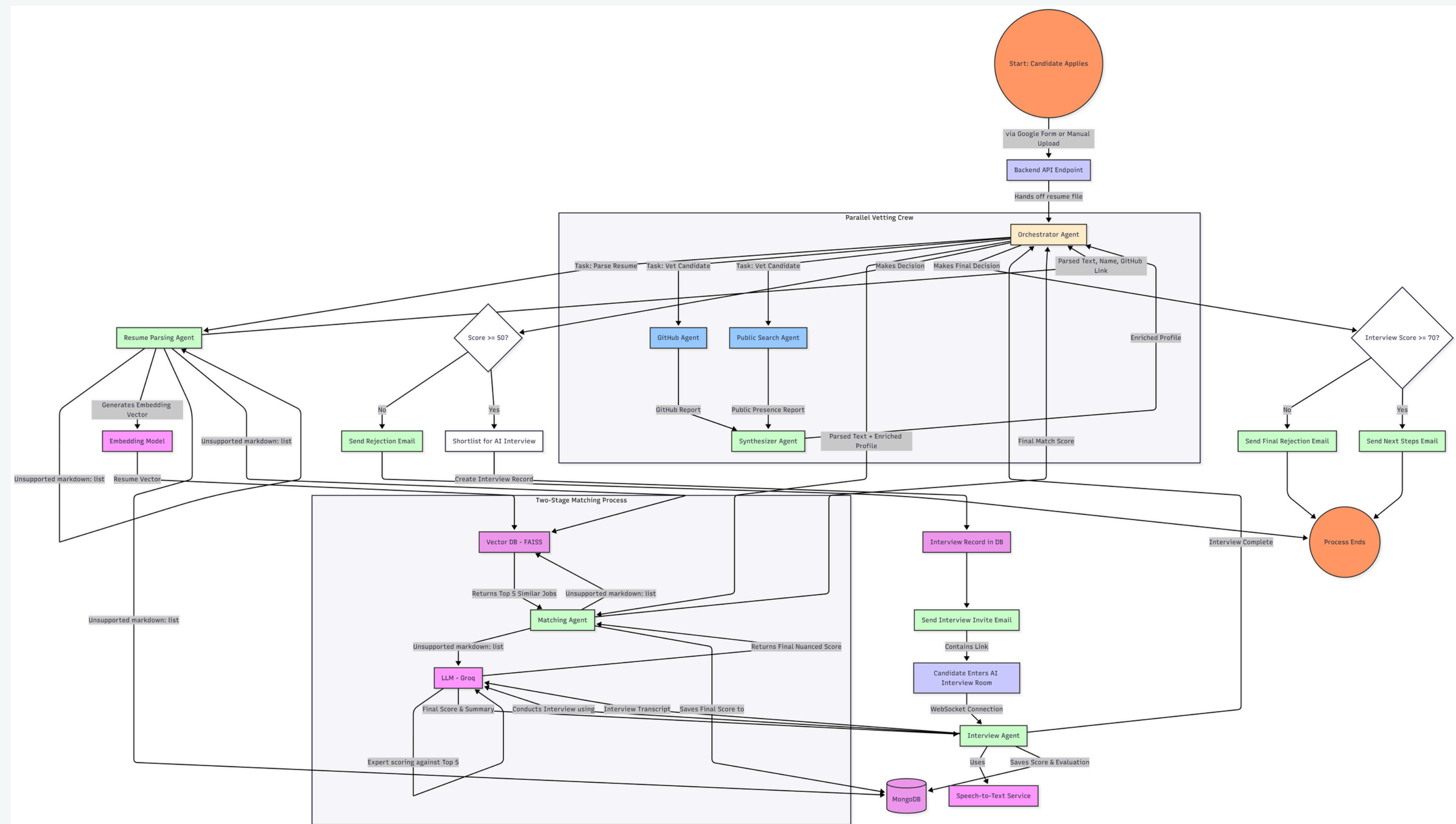


AI interview



Send final acceptance mail flowchart

System Architecture Overview



Quantitative Results

Metric	Before (Manual)	After (HireAI)	Improvement
Screening Time	3-4 days	3 hours	↓ 92%
Cost per Hire	₹8K-₹12K	₹3K-₹5K	↓ 55%
Drop-off Rate	25-30%	12%	↓ 60%
Match Accuracy	85%	92%	↑ 7%
Candidates in Parallel	1-2	20+	↑ 10×

Research & Validation

- Implemented semantic embedding scoring for candidate–job matching
- Added LLM fairness guardrails for bias detection
- Integrated hierarchical CrewAI memory for contextual persistence
- Conducted pilot with 5 HR datasets, achieving 92% match precision



References: LinkedIn Talent Report 2024, SHRM 2025, NASSCOM-BCG 2024, Grand View Research 2025

“Our validation shows HireAI can handle real-world hiring complexity.”

What Makes HireAI Unique

Feature	Traditional HR	HireAI
Voice Interview	✗	✓
Multi-Agent Orchestration	✗	✓
Fairness & Bias Checks	⚠	✓
Real-Time Analytics	✗	✓
LLM Memory Sharing	✗	✓

HireAI is the Atria University's first HR pipeline built entirely on an Agentic AI framework.



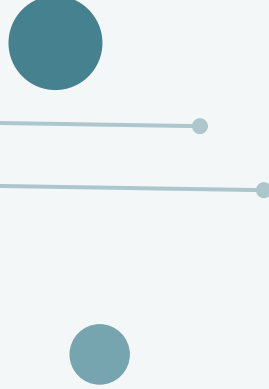
The Road Ahead: Evolving the Platform

- Human In The Loop Adding an approval step for a human recruiter
- REST APIs compatible with Zoho Recruit, Workable, Darwinbox
- Modular CrewAI orchestration for 100+ concurrent candidates
- HR dashboard for live analytics
- Multilingual input (English, Hindi, Kannada, Tamil)

Next Steps

Pilot with Bangalore-based SME
SaaS HR product
Expansion to Predictive HR Analytics

HireAI is production-ready — built for scale, fairness, and intelligence.



Team Roles

- **Deepesh: Project Lead & Lead AI Architect**
 - **Rayyan: Research , Backend & FastAPI Specialist**
 - **Vaibhav: Frontend & UI/UX Developer**
 - **Jewel: Agentic Workflow & Prompt Engineer**
- 



Live Implementation

DEMO



Conclusion

The Agentic AI Hiring System is more than just an automation tool; it is a strategic platform that makes recruitment faster, fairer, and smarter. By delegating the entire process to a crew of specialized AI agents, companies can mitigate legal risk, improve the quality of their hires, and gain a significant competitive advantage in the war for talent.



The background features abstract geometric elements in a teal color. In the top-left corner, two parallel lines extend diagonally downwards, each ending in a small dot. On the left side, there are three circles of varying sizes, stacked vertically. In the bottom-left corner, there is a single medium-sized circle. On the right side, two lines originate from the text area: one extends horizontally to the right with a dot at its end, and the other extends diagonally downwards and to the right, also ending in a dot.

Thank You!